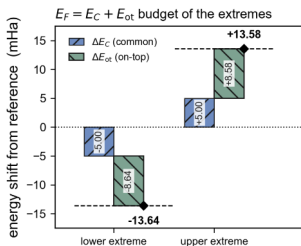
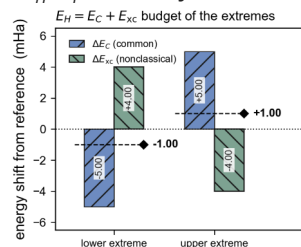
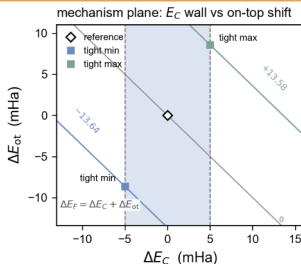
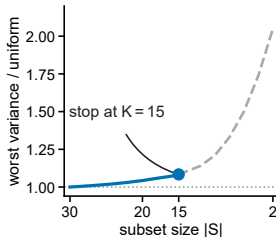


a **$E_H - E_F$ sensitivity mismatch**

same small 2-RDM moves ($\|\Delta D_2\|_F \leq 1.01\%$):
 $|\Delta E_H| \leq 1$ mHa vs $|\Delta E_F| \approx 13.6$ mHa

**b****frame selection**pilot: every frame $\times$ 500 shots

pool: 30 Haar O(8) frames

backward-greedy screen (pilot-only)keep $K = 15$, equal shots

($B - 15,000$)/15 new shots per kept frame
 kept pilots reused, dropped ones discarded

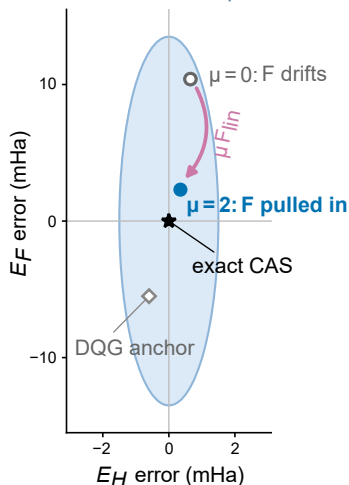
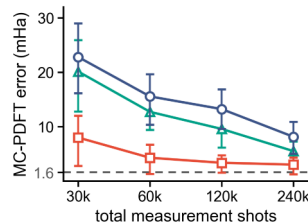
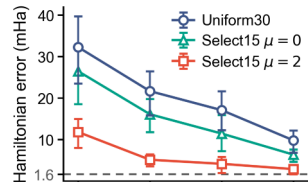
c **$\mu = 2$ ftPBE regularization**

band SDP on the 15 selected frames

$$\min_{D, \gamma, E^\pm} H + \mu F_{\text{lin}} + \lambda \text{Tr}(E^+ + E^-)$$

s.t. DQG 2-positivity, spin, contraction,
row band $\gamma \in A \text{vec}(D \pm \Delta)$

measurement-compatible set

**d****effect on H and F errors** N_2 , $R = 1.10$ Å N_2 dissociation, 120k shots